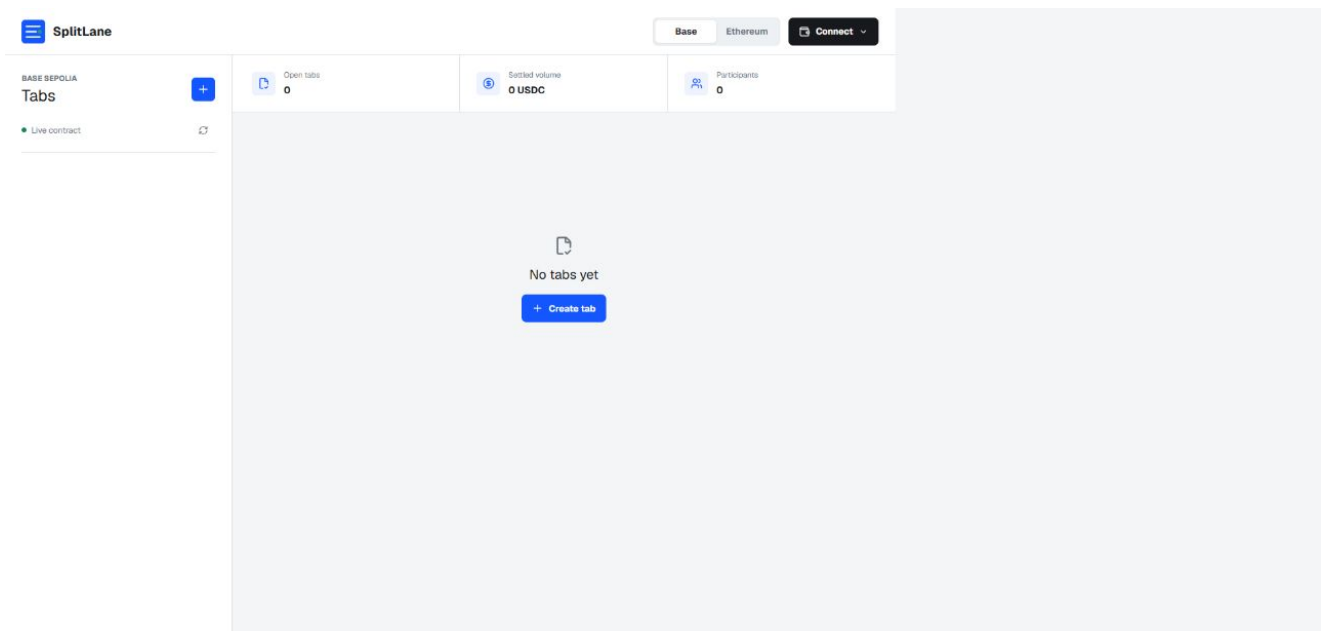


SplitLane

Non-custodial USDC group settlement with an Attestcoin proof lane

A recipient creates a group tab with exact shares. Each participant pays USDC directly to the recipient; the contract never pools funds or exposes an admin withdrawal path.



Live application: <https://splitlane.vercel.app>

Source: <https://github.com/liw38884-spec/splitlane>

Deployed on Base Sepolia and Ethereum Sepolia

Problem and Solution

The problem

- Group expenses are difficult to coordinate and reconcile across multiple wallets.
- Custodial split-payment systems create pooled-fund, recovery, and operator trust risks.
- Cross-chain payment evidence is often accepted through centralized APIs rather than verified infrastructure.

The SplitLane approach

- Store an immutable tab with recipient, title, participants, and exact USDC shares.
- Move each participant's USDC directly to the recipient using transferFrom.
- Let only the assigned participant settle their share and only the recipient close an unpaid tab.
- Use Attestcoin for Ethereum Sepolia receipt evidence on Creditcoin CC3 Testnet.

Target users

- Travel groups and shared households
- Small teams, communities, and event organizers
- Developers needing auditable cross-chain receipt workflows

Architecture and Attestcoin Integration

Layer	Responsibility
Next.js web app	Wallet connection, chain selection, tab creation, exact USDC approval, settlement, explorer links, and Base Builder Code attribution.
SplitLane Solidity contract	Immutable tab accounting and direct participant-to-recipient USDC transfer. No owner, upgrade, withdrawal, recovery, or payable path.
Attestcoin lane	Accepts Ethereum Sepolia only, inspects the source receipt, binds it to the configured SplitLane contract and selector, discovers chain configuration, and builds or verifies proof evidence on Creditcoin CC3 Testnet.

Why Attestcoin is core

- The proof lane rejects Base Sepolia rather than presenting unsupported cross-chain verification.
- Ethereum Sepolia receipt data is checked against the deployed SplitLane address and expected payShare function selector.
- Dry-run mode makes the evidence pipeline reproducible; verify mode uses an injected signer for CC3 submission.

Security Model

- Non-custodial settlement: SplitLane does not retain a pooled USDC balance.
- No privileged admin: there is no owner withdrawal or upgrade mechanism.
- No native-currency custody: payable calls and direct ETH transfers are rejected.
- Exact-share authorization: the frontend approves only the assigned USDC amount when needed.
- Reentrancy resistance: state-changing settlement paths use ReentrancyGuard and SafeERC20.
- Fail-closed proof lane: unsupported chain, wrong contract, wrong selector, missing signer, or failed source receipt stops verification.

Verification evidence

- 13 web tests plus TypeScript, ESLint, and production build.
- 15 Attestcoin lane tests plus TypeScript build.
- 30 Foundry tests, including fuzz tests and three invariant suites with 128,000 calls per invariant run.
- Both testnet deployments confirmed with successful receipts, deployed bytecode, and network-specific Circle USDC bindings.

Live Deployments and Roadmap

Network	Contract	Deployment transaction
Base Sepolia	0x1EE8dCEE85c5bD8bA8D21B599D08Acc3E80C0d6c	0xa2b4e8edc7c29865535adc6c6e221762efe2416231dc5807b4404596d6d2070e
Ethereum Sepolia	0x1EE8dCEE85c5bD8bA8D21B599D08Acc3E80C0d6c	0x1e0095247e4abfdb355e5e9e079572dd1fb0c4835db6d70abd28f1f077adfabd

Roadmap

- Complete connected-wallet create, approve, pay, close, and explorer-link QA on both testnets.
- Produce a live Ethereum Sepolia settlement receipt and complete Attestcoin CC3 verification.
- Add reminders, exports, team workspaces, and an API for recurring operational use.
- Commission an external security review before considering any mainnet deployment.

Public links

App: <https://splitlane.vercel.app>

Repository: <https://github.com/liw38884-spec/splitlane>

Builder Code: bc_qgmgm02h